

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Active-controlled, Open-label, Multicenter Study to Compare the Efficacy and Safety of MK-2870 Monotherapy Versus Treatment of Physician's Choice in Participants With Endometrial Cancer Who Have Received Prior Platinum-based Chemotherapy and Immunotherapy (MK-2870-005/ENGOT-en23/GOG-3095) **Short Title/Acronym:** TroFuse-005 / MK-2870-005 **Protocol Number:** MK-2870-005 **NCT Number:** NCT06132958 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Sacituzumab tirumotecan (MK-2870 / SKB264), a TROP2-directed antibody-drug conjugate (ADC) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate and compare if participants with previously treated endometrial cancer receiving sacituzumab tirumotecan live longer overall (Overall Survival, OS) and without their cancer getting worse (Progression-Free Survival, PFS) compared to those receiving the treatment of physician's choice (standard chemotherapy). **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DOR), safety/tolerability (including the number of participants who experience one or more adverse events), and patient-reported quality of life outcomes (measured via EORTC QLQ-C30).

2.2 Background & Rationale To address the unmet medical need in patients with advanced endometrial carcinoma or carcinosarcoma who have previously received platinum-based chemotherapy and immunotherapy (PD-1/PD-L1 inhibitors). For patients whose disease progresses after these standard therapies, further treatment options are limited. Sacituzumab tirumotecan targets the TROP2 antigen—which is highly expressed on many cancer cells—to deliver a targeted chemotherapy payload, potentially offering improved survival with manageable toxicity compared to traditional chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active-controlled (Treatment of Physician's Choice: Doxorubicin, Paclitaxel, or Nab-paclitaxel) **Blinding:** Open-label **Randomization:** Randomized (Parallel Assignment) **Status:** Active, not recruiting

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 710 participants **Number of Sites:** Multicenter, Global **Estimated Study Start:** 06-Dec-2023 **Estimated Primary Completion:** 10-Jan-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female, age ≥ 18 years. 2. Histologically-confirmed diagnosis of endometrial carcinoma or carcinosarcoma. 3. Radiographically evaluable disease, either measurable or nonmeasurable per Response Evaluation Criteria in Solid Tumors (RECIST 1.1), as assessed by Blinded Independent Central Review (BICR). 4. Has received prior platinum-based chemotherapy and anti-PD-1/anti-PD-L1 therapy, either separately or in combination.

4.2 Exclusion Criteria 1. Has neuroendocrine tumors or endometrial sarcoma, including stromal sarcoma, leiomyosarcoma, adenosarcoma, or other types of pure sarcomas. 2. Has a history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 3. Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease. 4. Has had a recurrence of endometrial carcinoma or carcinosarcoma more than >12 months after completing platinum-based therapy administered in the curative-intent setting without any additional platinum-based therapy received in the recurrent setting. Note: 1) If Immunotherapy-based treatment is administered in the recurrent setting, then platinum rechallenge is not required, regardless of the duration of the platinum-free interval from time of adjuvant therapy. 2) For Stage IVb disease, treatment that includes gynecological surgery followed by a platinum-based regimen is NOT considered curative-intent per protocol and does not require platinum rechallenge in the recurrent setting. 5. Has received more than 3 prior lines of therapy for endometrial carcinoma or carcinosarcoma. 6. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 7. Has received prior treatment with single-agent nonplatinum based chemotherapy in the third-line setting. 8. Has received prior treatment with a trophoblast cell surface antigen 2 (TROP2)-targeted antibody drug conjugate (ADC) (e.g., sacituzumab govitecan). 9. Has received prior treatment with a topoisomerase I inhibitor-containing ADC (e.g., sacituzumab govitecan or fam-trastuzumab deruxtecan-nxki). 10. Has previously received both single-agent paclitaxel and single-agent doxorubicin in any setting for prior treatment of endometrial cancer. 11. Requires recurrent drainage of effusions (e.g., pleural, ascitic, etc.) within 6 weeks before randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sacituzumab tirumotecan 4 mg/kg administered on Day 1 of each 14-day cycle. Premedication (diphenhydramine, H2 antagonist, acetaminophen, and dexamethasone) is administered prior to the first 4 infusions; H2 antagonist and dexamethasone are optional for subsequent infusions. *Active Comparator Arm:* Doxorubicin 60 mg/m² Q3W, OR Paclitaxel 80 mg/m² on Days 1, 8, and 15 of each 28-day cycle. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Continued until documented disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard premedications for infusion reactions and antiemetics. **Prohibited:** Concurrent investigational agents, additional systemic anti-cancer therapies, or previous exposure to TROP2-targeted ADCs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, BICR Baseline Scans, Labs
Baseline	Day 1	Randomization, First IV Infusion (MK-2870 or Chemotherapy)
Interim	Q2W, Q3W, or Q1W	Infusion visits dependent on assigned arm (14-day cycle for MK-2870, 21 or 28-day for chemo). Safety Labs, Efficacy Assessment, EORTC QLQ-C30
End of Study	Variable	End of Treatment Visit, Final Scans, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Progression-free Survival (PFS) per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 as Assessed by Blinded Independent Central Review (BICR): Defined as the time from randomization to the first documented disease progression per RECIST 1.1 based on BICR or death due to any cause, whichever occurs first. 2. Overall Survival (OS): Defined as the time from randomization to death due to any cause. **Measurement Method:** Radiographic imaging evaluated by Blinded Independent Central Review (BICR) and standard survival tracking.

7.2 Secondary & Exploratory Endpoints 1. Objective Response Rate (ORR) per RECIST 1.1 as Assessed by BICR 2. Duration of Response (DOR) per RECIST 1.1 as Assessed by BICR 3. Number of Participants Who Experience One or More Adverse Events (AEs) 4. Quality of Life/Health Status Changes via EORTC QLQ-C30 questionnaire

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for any untoward medical occurrences temporally associated with the use of study interventions. **Serious Adverse Events (SAEs):** Standard reporting timelines per international regulations; specific monitoring for infusion reactions, interstitial lung disease/pneumonitis, and severe ocular toxicities. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to review safety and efficacy interim data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy endpoints (PFS/OS). Safety Analysis Set for patients receiving at least one dose of study medication. **Sample Size Justification:** Approximately 710 participants globally, powered to detect superiority in PFS and OS for MK-2870 vs. Treatment of Physician's Choice. **Handling of Missing Data:** Standard oncology censoring rules applied for PFS and OS; linear transformation and appropriate imputation for QoL measures.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of clinical sites, strict oversight of IV administration protocols. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with centralized radiological review via BICR to ensure unbiased efficacy assessments.

11. Study Identifiers & Governance

Primary ID: MK-2870-005 **Registry Number:** NCT06132958 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** Sacituzumab Tirumotecan (MK-2870) in Post Platinum and Post Immunotherapy Endometrial Cancer (MK-2870-005) **Official Regulatory Title:** A Phase 3, Randomized, Active-controlled, Open-label, Multicenter Study to Compare the Efficacy and Safety of MK-2870 Monotherapy Versus Treatment of Physician's Choice in Participants With Endometrial Cancer Who Have Received Prior Platinum-based Chemotherapy and Immunotherapy (MK-2870-005/ENGOT-en23/GOG-3095)

12. Clinical Framework (Endometrial Cancer Specific)

Primary Condition: Endometrial Cancer (Endometrial Carcinoma or Carcinosarcoma) **Diagnosis Threshold:** Advanced/metastatic disease

progression following platinum-based chemotherapy and PD-1/PD-L1 immunotherapy **Medical Methodology:** TROP2-targeted Antibody-Drug Conjugate (ADC) therapy vs. standard single-agent chemotherapy (Doxorubicin, Paclitaxel, Nab-paclitaxel) **Optimization Target:** Delayed disease progression (PFS) and extended Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Systemic oncological treatment for advanced gynecological malignancy. **Education Protocol (HCP):** Specialized training for ADC handling, IV infusion protocols, and management of TROP2-ADC specific toxicities (e.g., ocular, lung). **Education Protocol (Patient):** Informed consent detailing required premedication, side effect management, and reporting of severe symptoms. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of all tumor imaging.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 years (Estimated primary completion January 2028) **Onsite Visit Cadence:** Every 14 days for MK-2870 arm; every 21 or 28 days for comparator arms **Remote Monitoring Frequency:** Routine adverse event and survival tracking. **Checklist Review Interval:** Regular tumor assessments (e.g., Q8W to Q12W) to determine RECIST 1.1 progression.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				